



MANIPAL ACADEMY OF HIGHER EDUCATION

**THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS -
JAN/FEB 2023**

**SUBJECT : CSE 2152 - DATA STRUCTURES AND APPLICATIONS
(MAKEUP)**

DATA STRUCTURES AND APPLICATIONS [CSE 2152]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

1A) Solve the problem of incrementing a date using a C program. Write a function called increment that accepts a pointer to a date structure with three fields: day, month, year (all are of type integers). The function increments the date by 1 day and returns the new date. If the date is the last day of the month, the month field must also be changed. If the month is December, the value of the year field must also be changed when the day is 31. (4)
In addition, the function should take care while incrementing the day in the month of February (consider leap year and non-leap year). Test this function from main function by creating a pointer to a date structure using dynamic memory allocation.

1B) Create a C program that reads an integer from the keyboard and then calls recursive function to print it out in reverse. For example, if the user enters 4762, it prints 2674. (3)

1C) Imagine, we want to read a list of numbers, and each time we read a negative number we must backtrack (that is, to retrace its steps) and print the five numbers that come before the negative number and then discard the negative number. If there are fewer than five elements, print an error message and stop the program. After printing the five numbers, resume reading the data and this process should continue. Solve this problem of backtracking using a suitable data structure. (3)

2A) Outline an algorithm to convert a postfix expression to prefix expression. Apply the same algorithm to convert the expression $A B C - D * + E * F +$ to its prefix form by showing the steps involved as per the table provided: (4)

Current symbol scanned	Action Taken (push/pop etc.)	Stack Content

2B) Apply the algorithm for evaluation of postfix expression and evaluate the following expression. Show the contents of the given table after every operation. (Consider: A=8; B=7; C=6; D=5; E=4; F=3; G=2; H=1)
Expression: $A B C / D E F * + G / + H * -$ (3)

Symbol scanned from postfix expression	Action Taken (Evaluate/Push/Pop etc.)	Stack Content

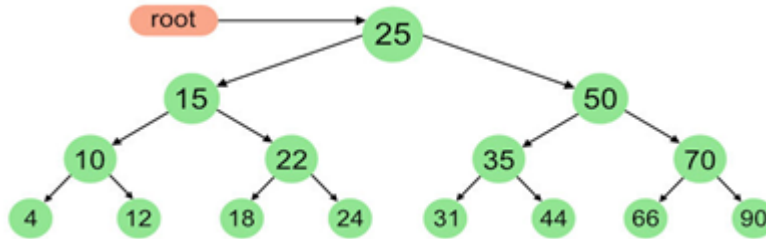
2C) Write C functions to implement insert and delete operations on linear queue of integers. Write a main function to invoke the above functions and display the contents of queue after every operation. (3)

3A) Write functions for stack operations push() and pop() using Singly Linked List. Write the structure of the node as well. (3)

3B) Assuming a Singly Linked List is passed as a parameter, write the necessary structure and a function sort() to sort the given list in ascending order and display the same. (3)

3C) Create a Doubly linked list of 3 nodes with integer data. Obtain sum of data by traversing from left to right and difference by traversing from right to left. Print the list and output. (4)

4A) Write Inorder, preorder traversal and recursive c program for inorder for the given tree.

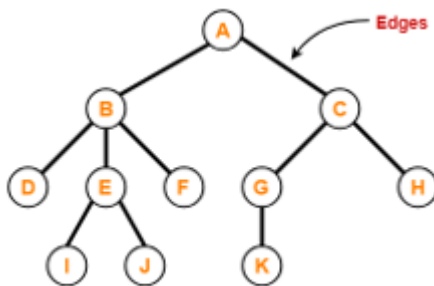


(1+1+1 = 3 marks)

4B) Write a function for the insertion of an element in a threaded binary tree. (3)

4C) Write the deletion function of the Binary search tree. Make Binary search tree for the given element, show all the steps in details: 10,15,12,5,4,20,8,7,14,13. (2+2 = 4 marks) (4)

5A) Write down the root, level, depth, and subtree that are present in the given tree:



($\frac{1}{2}$ mark $\times$ 4 = 2 marks)]

5B) Consider the following sparse matrix and write how to represent this sparse matrix in linked list and array.

0 0 3 0 4

0 0 5 7 0

0 0 0 0 0

0 2 6 0 0

($1\frac{1}{2}+1\frac{1}{2}$ = 3 marks)

5C) Write an algorithm to check whether given expression is valid or not using stack. With an example, explain different cases in which these expression validations is checked. (5)

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